



- 7 A discrete random variable X takes values $r = 0, 1, 2$ with probabilities $P(X = r)$ as given in the following table.

r	0	1	2
$P(X=r)$	a	$2a$	b

- (a) Write down the probability generating function of X , and use it to find an expression for $E(X)$ in terms of a and b . [2]

[illegible]

- (b) Show that $\text{Var}(X) = 2b + 2(a+b)(1-2a-2b)$. [3]

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The random variable Y is defined by $Y = X_1 + X_2 + X_3 + \dots + X_{10}$ where $X_1, X_2, X_3, \dots, X_{10}$ are ten independent observations of X .

- (c) Using the probability generating function of Y , and your answer to part (a), show that $E(Y) = 10E(X)$. [3]

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- (d) For the case $b = 0$, define fully the distribution of Y . [2]

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